BADMINTON PERFORMANCE REPORT

Statistical Analysis and Performance Prediction

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[Project Motto]

Contents

1 Introduction

This report presents a systematic analysis of badminton match records across three formats: Singles (1v1), Doubles (2v2), and Mixed Doubles (Mix).

The recorded scores are organised chronologically and examined using statistical methods to understand performance, consistency, and changes in scoring behaviour.

1.1 Objectives

The report has three primary objectives.

1.1.1 Performance Analysis

The first objective is to analyse the recorded match results using descriptive statistics.

This includes calculating averages, standard deviations, total scores, and net results. These quantities help describe scoring performance and the degree of variation across matches.

1.1.2 Next-Game Prediction

The second objective is to develop a statistical model that estimates the expected score in the next game.

A least-squares regression model will be considered using the historical match records. The objective is to identify whether the recorded scoring pattern provides a useful basis for estimating a future result.

The prediction represents a statistical estimate, not a guaranteed outcome.

1.1.3 Interpretation

The final objective is to interpret the calculated statistics and graphical results.

The analysis will examine differences between match formats, scoring

consistency, trends, and the limitations of making predictions from a finite set of observations.

1.2 Match Formats

- **Singles (1v1):** One player competes against one opponent.
- **Doubles (2v2):** Two players compete against two opponents.
- **Mixed Doubles (Mix):** Each team consists of one male and one female player.

1.3 Scoring Convention

The recorded scores are interpreted as follows:

- **Win:** Points scored by our team.
- **Lose:** Points scored by the opponent.

The net result is defined as

$$N = W - L$$

where W represents the Win score and L represents the Lose score.

A positive net result indicates a higher score for our team, while a negative value indicates a higher score for the opponent.

1.4 Treatment of Missing Entries

If one score is missing while the other is recorded, the missing score is considered zero.

If both scores are absent for a particular match format on a given date, that entry is considered an absent match and is excluded from the corresponding table and calculations.

1.5 Mottos

Insert the supplied mottos here.

2 Match Records

This section presents the recorded badminton match results, organised according to the three match formats: Singles (1v1), Doubles (2v2), and Mixed Doubles (Mix).

Each table records the date, points scored by our team, points scored by the opponent, and the net result.

The net result is calculated as

$$\text{Net Result} = \text{Win} - \text{Lose}$$

A positive value indicates that our team scored more points, while a negative value indicates that the opponent scored more points.

Entries with both scores absent for a particular format are excluded. If only one score is absent, the missing value is treated as zero.

2.1 Singles (1v1)

Singles matches involve one player from each team. The following table presents the recorded Singles scores in chronological order.

Table 1: Singles Match Records (1v1)

No.	Date	Win	Lose	Net
1	26-05-2026	3	2	+1
2	28-05-2026	1	1	0
3	30-05-2026	5	0	+5
4	31-05-2026	1	0	+1
5	02-06-2026	8	0	+8
6	03-06-2026	1	0	+1
7	04-06-2026	0	0	0
8	05-06-2026	3	0	+3
9	06-06-2026	0	1	-1

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Table 1 – continued

No.	Date	Win	Lose	Net
10	07-06-2026	3	0	+3
11	08-06-2026	1	0	+1
12	09-06-2026	3	0	+3
13	11-06-2026	2	0	+2
14	17-06-2026	4	0	+4
15	18-06-2026	4	1	+3
16	19-06-2026	1	0	+1
17	20-06-2026	4	0	+4
18	21-06-2026	2	0	+2
19	22-06-2026	1	0	+1
20	25-06-2026	5	0	+5
21	05-07-2026	2	1	+1
22	07-07-2026	1	0	+1
23	10-07-2026	1	0	+1
24	11-07-2026	0	1	-1
25	24-07-2026	6	0	+6
26	25-07-2026	1	0	+1
27	26-07-2026	1	3	-2
28	29-07-2026	1	0	+1
29	31-07-2026	3	0	+3
30	04-08-2026	2	0	+2
31	05-08-2026	2	1	+1
32	07-08-2026	1	0	+1
33	10-08-2026	6	0	+6
34	11-08-2026	2	1	+1
35	12-08-2026	2	1	+1
36	13-08-2026	1	0	+1
37	14-08-2026	3	0	+3
38	15-08-2026	1	0	+1
39	17-08-2026	5	1	+4
40	18-08-2026	3	0	+3
41	19-08-2026	6	0	+6
42	20-08-2026	2	0	+2

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Table 1 – continued

No.	Date	Win	Lose	Net
43	22-08-2026	0	1	-1
44	23-08-2026	1	0	+1
45	26-08-2026	3	0	+3
46	27-08-2026	3	0	+3
47	28-08-2026	3	2	+1
48	30-08-2026	1	2	-1
49	31-08-2026	2	2	0
50	01-09-2026	0	1	-1
51	02-09-2026	1	1	0
52	03-09-2026	5	0	+5
53	06-09-2026	1	4	-3
54	07-09-2026	2	1	+1
55	09-09-2026	1	1	0
56	10-09-2026	2	3	-1
57	11-09-2026	2	0	+2
58	16-09-2026	2	1	+1
59	17-09-2026	1	1	0
60	18-09-2026	5	0	+5
61	21-09-2026	2	0	+2
62	23-09-2026	2	0	+2
63	24-09-2026	2	0	+2
64	27-09-2026	2	0	+2
65	28-09-2026	2	2	0
66	29-09-2026	3	0	+3
67	05-10-2026	1	2	-1
68	07-10-2026	2	0	+2
69	09-10-2026	1	1	0
70	13-10-2026	3	0	+3
71	14-10-2026	5	0	+5
72	27-10-2026	4	0	+4
73	28-10-2026	6	0	+6
74	29-10-2026	4	0	+4
75	02-11-2026	3	0	+3

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Table 1 – continued

No.	Date	Win	Lose	Net
76	03-11-2026	2	0	+2
77	04-11-2026	2	0	+2
78	05-11-2026	2	2	0

2.2 Doubles (2v2)

Doubles matches involve two players on each team. The following table presents the recorded Doubles scores in chronological order.

Table 2: Doubles Match Records (2v2)

No.	Date	Win	Lose	Net
1	26-05-2026	4	5	-1
2	27-05-2026	6	4	+2
3	28-05-2026	6	6	0
4	29-05-2026	6	2	+4
5	30-05-2026	0	3	-3
6	31-05-2026	5	3	+2
7	02-06-2026	4	5	-1
8	03-06-2026	5	1	+4
9	04-06-2026	15	3	+12
10	05-06-2026	8	5	+3
11	06-06-2026	6	0	+6
12	07-06-2026	5	4	+1
13	08-06-2026	3	7	-4
14	09-06-2026	2	3	-1
15	10-06-2026	4	6	-2
16	11-06-2026	1	2	-1
17	15-06-2026	1	3	-2
18	17-06-2026	1	2	-1
19	18-06-2026	2	3	-1
20	19-06-2026	2	3	-1
21	20-06-2026	2	3	-1
22	21-06-2026	6	2	+4

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Table 2 – continued

No.	Date	Win	Lose	Net
23	22-06-2026	4	5	-1
24	23-06-2026	2	4	-2
25	24-06-2026	2	2	0
26	25-06-2026	3	1	+2
27	01-07-2026	1	3	-2
28	05-07-2026	2	5	-3
29	06-07-2026	2	1	+1
30	07-07-2026	0	4	-4
31	08-07-2026	4	6	-2
32	09-07-2026	5	3	+2
33	10-07-2026	3	4	-1
34	11-07-2026	3	2	+1
35	12-07-2026	4	4	0
36	23-07-2026	3	4	-1
37	24-07-2026	2	4	-2
38	25-07-2026	3	2	+1
39	26-07-2026	6	5	+1
40	27-07-2026	1	8	-7
41	29-07-2026	2	2	0
42	30-07-2026	4	4	0
43	31-07-2026	4	1	+3
44	03-08-2026	2	2	0
45	04-08-2026	6	5	+1
46	05-08-2026	2	1	+1
47	06-08-2026	2	1	+1
48	07-08-2026	3	2	+1
49	08-08-2026	4	1	+3
50	09-08-2026	1	4	-3
51	10-08-2026	7	3	+4
52	11-08-2026	3	5	-2
53	12-08-2026	4	6	-2
54	13-08-2026	12	2	+10
55	14-08-2026	2	1	+1

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Table 2 – continued

No.	Date	Win	Lose	Net
56	15-08-2026	3	3	0
57	16-08-2026	3	3	0
58	17-08-2026	3	2	+1
59	18-08-2026	2	0	+2
60	19-08-2026	4	1	+3
61	20-08-2026	3	1	+2
62	21-08-2026	5	0	+5
63	22-08-2026	3	3	0
64	23-08-2026	5	6	-1
65	24-08-2026	6	5	+1
66	25-08-2026	9	6	+3
67	26-08-2026	1	5	-4
68	27-08-2026	9	2	+7
69	28-08-2026	3	0	+3
70	30-08-2026	2	4	-2
71	31-08-2026	3	1	+2
72	01-09-2026	4	1	+3
73	02-09-2026	5	6	-1
74	03-09-2026	4	3	+1
75	06-09-2026	5	0	+5
76	07-09-2026	1	2	-1
77	09-09-2026	2	8	-6
78	11-09-2026	3	1	+2
79	16-09-2026	2	0	+2
80	17-09-2026	2	0	+2
81	18-09-2026	1	4	-3
82	19-09-2026	0	2	-2
83	21-09-2026	0	4	-4
84	23-09-2026	1	3	-2
85	24-09-2026	2	3	-1
86	27-09-2026	1	2	-1
87	28-09-2026	2	2	0
88	29-09-2026	2	0	+2

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Table 2 – continued

No.	Date	Win	Lose	Net
89	07-10-2026	5	0	+5
90	08-10-2026	2	6	-4
91	09-10-2026	1	3	-2
92	12-10-2026	6	6	0
93	13-10-2026	10	4	+6
94	14-10-2026	2	2	0
95	15-10-2026	4	1	+3
96	26-10-2026	3	2	+1
97	27-10-2026	4	1	+3
98	28-10-2026	2	0	+2
99	29-10-2026	2	1	+1
100	02-11-2026	2	3	-1
101	03-11-2026	2	5	-3
102	04-11-2026	3	2	+1
103	05-11-2026	0	7	-7

2.3 Mixed Doubles (Mix)

Mixed Doubles matches involve one male and one female player on each team.

The following table presents the recorded Mixed Doubles scores in chronological order.

Table 3: Mixed Doubles Match Records (Mix)

No.	Date	Win	Lose	Net
1	27-05-2026	0	3	-3
2	28-05-2026	1	0	+1
3	31-05-2026	1	1	0
4	03-06-2026	1	0	+1
5	04-06-2026	2	0	+2
6	06-06-2026	2	0	+2
7	10-06-2026	2	4	-2
8	18-06-2026	1	0	+1
9	19-06-2026	1	2	-1
10	22-06-2026	1	0	+1

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Table 3 – continued

No.	Date	Win	Lose	Net
11	24-06-2026	1	0	+1
12	08-07-2026	0	1	-1
13	09-07-2026	2	0	+2
14	10-07-2026	1	0	+1
15	25-07-2026	1	0	+1
16	26-07-2026	0	2	-2
17	27-07-2026	0	3	-3
18	30-07-2026	2	3	-1
19	04-08-2026	0	1	-1
20	07-08-2026	1	0	+1
21	08-08-2026	1	0	+1
22	10-08-2026	1	1	0
23	17-08-2026	2	1	+1
24	19-08-2026	2	0	+2
25	21-08-2026	2	0	+2
26	27-08-2026	1	0	+1
27	31-08-2026	4	0	+4
28	09-09-2026	1	0	+1
29	19-09-2026	1	0	+1
30	21-09-2026	2	0	+2
31	23-09-2026	1	0	+1
32	26-10-2026	1	0	+1
33	03-11-2026	2	0	+2

3 Statistical Analysis

This section analyses the recorded badminton match results using statistical methods. The objective is to understand the team's performance across Singles, Doubles, and Mixed Doubles formats.

The analysis considers the points scored by our team, the points scored by the opponents, and the difference between the two. Descriptive statistics are used to identify general performance patterns and variations in the recorded results.

3.1 Performance Parameters

The following parameters are used throughout the analysis.

- **Wins:** Points scored by our team.
- **Losses:** Points scored by the opposing team.
- **Net Result:** Difference between points scored by our team and the opponent.

For a match record, the net result is calculated as

$$R_i = W_i - L_i$$

where:

- W_i represents the points scored by our team in the i-th match.
- L_i represents the points scored by the opponent in the i-th match.
- R_i represents the net result.

A positive net result indicates that our team scored more points, while a negative value indicates that the opponent scored more points.

3.2 Descriptive Statistics

Descriptive statistics provide a summary of the recorded performance in each match format.

The following statistical measures are considered.

3.2.1 Mean

The arithmetic mean represents the average number of points scored per recorded match.

For the points scored by our team,

$$\overline{W} = \frac{1}{n}\sum_{i=1}^{n} W_i$$

Similarly, the average points scored by the opponent are given by

$$\overline{L} = \frac{1}{n}\sum_{i=1}^{n} L_i$$

The average net result is

$$\overline{R} = \frac{1}{n}\sum_{i=1}^{n} R_i$$

where n is the number of valid records for the respective match format.

3.2.2 Standard Deviation

The standard deviation measures the variation of individual match results around their mean.

For our team's points,

$$\sigma_W = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(W_i - \overline{W})^2}$$

A lower standard deviation indicates that the recorded scores are relatively close to the average. A higher value indicates greater variation between matches.

The same calculation can be applied to the opponent's scores and the net results.

3.2.3 Total Points

The total points scored by both teams are calculated as

$$W_{\text{total}} = \sum_{i=1}^{n} W_i$$

$$L_{\text{total}} = \sum_{i=1}^{n} L_i$$

The cumulative net result is

$$R_{\text{total}} = W_{\text{total}} - L_{\text{total}}$$

These values provide an overall comparison of scoring performance within each match format.

3.3 Comparison Across Match Formats

The three match formats are analysed independently to understand how performance differs between Singles, Doubles, and Mixed Doubles.

The comparison considers:

1. Number of recorded matches.
2. Total points scored by our team.
3. Total points scored by opponents.
4. Average points scored per match.
5. Average net result.
6. Standard deviation of net results.

The comparison can be summarised using the following table.

Table 4: Summary of Statistical Measures

Parameter	**Singles**	**Doubles**	**Mixed**
Number of Matches	–	–	–
Total Wins	–	–	–
Total Losses	–	–	–
Average Wins	–	–	–
Average Losses	–	–	–
Average Net Result	–	–	–
Std. Deviation	–	–	–

The table allows the three formats to be compared using consistent statistical measures.

3.4 Date-wise Performance Analysis

Date-wise analysis is used to identify the variation in performance over the observation period.

For each date, the net result is calculated using

$$R_d = W_d - L_d$$

where W_d and L_d represent the total points recorded for our team and the opponent on date d, respectively, for the selected match format.

The recorded dates can then be ranked according to their net results.

A higher positive value indicates a stronger scoring advantage, whereas a negative value indicates a scoring disadvantage.

This ranking helps identify dates with comparatively strong and weak performances.

3.5 Graphical Analysis

Graphs are used to make the statistical patterns easier to interpret.

The following visualisations are considered for each match format.

3.5.1 Points Scored Over Time

A time-series plot is used to represent the points scored by our team and the opponent across the recorded dates.

The horizontal axis represents the date, while the vertical axis represents the number of points.

This plot helps identify changes in scoring performance over the observation period.

3.5.2 Net Result Over Time

A separate plot represents the net result for each recorded date.

A horizontal reference line at zero separates positive and negative net results.

The average net result can also be represented as a horizontal reference line to compare individual dates against the overall average.

3.5.3 Distribution of Net Results

A distribution plot can be used to understand how frequently different net-result values occur.

This helps determine whether the results are concentrated around the average or spread across a wide range of values.

3.6 Interpretation

The statistical measures and graphical representations provide complementary views of performance.

The mean describes the average scoring performance, while the standard deviation indicates how much individual results vary.

The cumulative net result gives an overall scoring comparison, whereas date-wise rankings identify individual periods of relatively strong or weak performance.

Together, these measures establish a statistical basis for comparing the three match formats and provide the historical data required for the next-game prediction model discussed in the following section.

4 Next-Game Prediction

The statistical analysis of historical match records provides a basis for estimating the scoring performance of the next game. This section develops a prediction model using the recorded scores from previous matches.

The objective is to estimate the points likely to be scored by our team and the opponent in the next game. The prediction is performed independently for Singles, Doubles, and Mixed Doubles.

4.1 Prediction Objective

The prediction model aims to estimate three quantities:

1. Expected points scored by our team.
2. Expected points scored by the opponent.
3. Expected net result of the next game.

The predicted net result provides an indication of the expected scoring advantage.

A positive predicted net result suggests that our team may score more points, while a negative value suggests that the opponent may score more points.

These predictions are estimates based on historical data and do not guarantee the outcome of a future game.

4.2 Least-Squares Linear Regression

Least-squares linear regression is used to model the relationship between the chronological match index and the recorded scores.

Let the historical matches be represented by

$$(x_i, W_i), \qquad i = 1, 2, \ldots, n$$

where:

- x_i is the chronological index of the i-th recorded match.
- W_i is the points scored by our team.
- n is the number of valid match records.

A linear model is expressed as

$$\widehat{W}(x) = a_W + b_W x$$

where a_W is the intercept and b_W is the slope.

The slope and intercept are determined by minimizing the sum of squared prediction errors:

$$S_W = \sum_{i=1}^{n} [W_i - (a_W + b_W x_i)]^2$$

The least-squares estimates are

$$b_W = \frac{n \sum x_i W_i - (\sum x_i)(\sum W_i)}{n \sum x_i^2 - (\sum x_i)^2}$$

and

$$a_W = \overline{W} - b_W \overline{x}$$

The same procedure is applied independently to the opponent's scores.

Thus, the opponent's scoring model is

$$\widehat{L}(x) = a_L + b_L x$$

where L_i represents the points scored by the opponent.

4.3 Predicting the Next Game

If n valid records are available for a particular match format, the next chronological index is

$$x_{\text{next}} = n + 1$$

The predicted points scored by our team are

$$\widehat{W}_{\text{next}} = a_W + b_W(n+1)$$

Similarly, the predicted points scored by the opponent are

$$\widehat{L}_{\text{next}} = a_L + b_L(n+1)$$

The predicted net result is then calculated as

$$\widehat{R}_{\text{next}} = \widehat{W}_{\text{next}} - \widehat{L}_{\text{next}}$$

This gives an estimated scoring comparison for the next game.

4.4 Separate Models for Match Formats

The prediction process is performed independently for all three match formats.

4.4.1 Singles Prediction

The Singles model uses only historical Singles records. Separate regression equations are fitted for our team's points and the opponent's points.

The resulting equations are used to estimate the next Singles score.

4.4.2 Doubles Prediction

The Doubles model uses only historical Doubles records. This ensures that the prediction is based on the scoring pattern observed in Doubles matches.

Separate equations are fitted for both teams to estimate the next Doubles score.

4.4.3 Mixed Doubles Prediction

The Mixed Doubles model uses only historical Mixed Doubles records. The same least-squares procedure is applied to estimate the next Mixed Doubles score.

Using separate models avoids combining scoring patterns from different match formats.

4.5 Prediction Summary

The final predictions can be presented in the following table after evaluating the regression equations.

Table 5: Estimated Scores for the Next Game

Format	Our Points	Opponent Points	Net Result
Singles	–	–	–
Doubles	–	–	–
Mixed Doubles	–	–	–

The predicted scores are continuous numerical estimates. Since actual match scores are integers, the estimates may be rounded to the nearest non-negative integer for presentation.

4.6 Limitations of the Model

The prediction model is based on historical scoring records and assumes that the observed linear trend provides useful information about future performance.

However, several factors can influence the actual result:

- Changes in the players participating in a match.
- Differences in opponent strength.
- Player fatigue and physical condition.
- Variations in playing conditions.
- The relatively small number of records available for some match formats.

The model does not explicitly account for these factors. Therefore, its output should be interpreted as a statistical estimate rather than a guaranteed match result.

4.7 Conclusion

Least-squares linear regression provides a simple method for estimating the next game's scoring performance from historical records.

By fitting separate models for Singles, Doubles, and Mixed Doubles, the prediction process preserves the differences between the three match formats.

The predicted scores and net results provide a quantitative estimate of future scoring performance and complement the descriptive statistical analysis presented in the previous section.